THE PHYSICAL AI CO-DESIGN MATRIX

Crossing Cognitive Agency (Chapter 3 Rows) with Physical Reality (Chapter 2 Columns)

	1. TIME & FRESHNESS Δt_{wall}, P99 Tails	**2. INERTIA & STOPPING** $p = mv$, d_{stop} Envelopes	**3. JERK & ACTUATION** $d\tau/dt$, Gearbox Tooth Shear	**4. SILICON & UMA BUS** DMA Saturation, Zero Heap
STAGE 1: PERCEIVE VLA Spatial Tokens 3D Affordances SE(3) → CHAPTER 4	**Exposure Latency** Sensor photon collection Hardware PTP timestamp cross-sensor sync	**Motion Blur Smear** $\Delta x_{blur} = v \cdot t_{exp}$ Diffuse gradient destroys millimeter grasp precision	**Rolling Shutter Shear** Sequential line readout deforms geometric frames under angular velocity ω	**DMA Ingestion Tax** 1.5 GB/s video write bursts saturate UMA crossbar; zero-copy ring buffers
STAGE 2: REMEMBER Latent World Models SE(3) Transform Trees → CHAPTER 5	**Belief Age Decay** Unobserved state aging State validity leases expire after Δt_{TTL}	**Occlusion Drift** $\sigma(t) = \sigma_0 + v_{target} \cdot \Delta t$ Expanding spatial volume enforces speed limits	**Winding Thermal Drift** Stator I^2R temperature rise derates maximum available actuator torque	**Frame Graph Lookups** SE(3) tf2 transform graph traversal latency must be cached in shared SRAM
STAGE 3: REASON Multimodal VLMs Intent Leases L_intent → CHAPTER 6	**Slow Deliberation (1 Hz)** High compute latency decoupled from control; operates as proposal service	**Kinematic Reach Bounds** Algebraic workspace filter validates 3D targets before trajectory planning	**Speed Clamping** Intent lease enforces maximum operational velocity clamp $\lVert v \rVert \leq v_{max}$	**Expiring Leases (L_intent)** Bounded target volume B, hard TTL expiration; fall back to safe hold
STAGE 4: PLAN Action Chunking (ACT) Diffusion Policies → CHAPTER 7	**Delay Amortization** Chunk horizon H=16 amortizes 50ms compute lag: $t_{amort} = t_{infer} / H$	**Receding Horizon** Overlapping chunk temporal ensembling prevents end-of-chunk stall	**C^2 Jerk Continuity** Quintic polynomial splines eliminate infinite torque rate & gearbox tooth shear	**SRAM Chunk Ring Buffer** Atomic lock-free queue bridges 20 Hz MPU planner to 1 kHz MCU interpolator
STAGE 5: EXECUTE 1 kHz Real-Time Reflex Control Barrier Functions → CHAPTER 8	**1000 Hz Determinism** Non-negotiable 1ms tick; jitter bounded to < 15μs; zero OS paging stalls	**Dynamic Stopping (d_stop)** Continuous clearance check: $d_{stop} \leq d_{clear}$; IEC 60204-1 dynamic halt	**Safe Torque Off (STO)** Hardware power relays & gate enable lines under exclusive MCU authority	**Bare-Metal CBF QP Solver** Zero dynamic heap malloc; projects proposals onto forward safe set: $h(x) \geq 0$